

JavaScript + Node.js Learning Roadmap

Prepared for a Node.js Developer with 3 Years Experience

Goal: Prepare for job switch by mastering JavaScript fundamentals and progressing into advanced Node.js topics.

JavaScript Topics (Completed/In Progress)

1. Core JavaScript Concepts

- Type Coercion (== vs ===), Primitive vs Reference Types
- Variable Scoping (var, let, const)
- Hoisting, Closures & Lexical Scope
- Temporal Dead Zone, Callback functions
- Promises & Async/Await, Microtasks vs Macrotasks
- Event Loop, Call Stack, setTimeout / setInterval
- Debounce & Throttle (concept + implementation)

2. Objects & Arrays

- Object creation (literal & constructor)
- Shallow vs Deep Copy (Object.assign, spread, JSON.parse, structuredClone)
- Destructuring, Optional chaining, Nullish coalescing
- Array methods (map, filter, reduce, some, every, find)
- Spread & Rest operators, Removing duplicates, Flattening

3. Functions

- Function Declarations vs Expressions, Arrow functions

- First-class & Higher-order functions
- Currying, bind/call/apply, IIFE, Function scope vs block scope

4. Asynchronous JavaScript

- Promise chaining, Promise.all/race/allSettled
- Custom Promise.all polyfill, Async/Await with try...catch
- Retry logic, Sequential vs Parallel async

5. ES6+ Features

- Template literals, Classes & Inheritance, Default Parameters
- Module system (import/export), Symbols, Set, Map, Destructuring

6. Error Handling

- try...catch...finally, Async error handling, Custom error classes
- Global error handling (onerror, unhandledrejection)

7. JavaScript Object Model

- Prototypes & chaining, Object.create, proto vs prototype
- Getters & Setters, freeze() vs seal(), hasOwnProperty vs in

8. Execution & Memory

- Execution Context, Lexical Environment
- Call Stack, Memory leaks, Garbage Collection, Tail Call Optimization

9. Functional Programming

- Pure functions, Immutability, Currying, Composition, Recursion

10. Tooling / Ecosystem

- Babel, Webpack/Vite, NPM vs Yarn, Scripts
- ESLint & Prettier, Tree shaking, .env usage

Node.js Roadmap (Upcoming)

1. Core Node.js

- Event-driven architecture, Node.js process model
- Modules (CommonJS & ESM), File system module (fs)
- Path, OS, Buffer, Events, Streams

2. Asynchronous Programming in Node

- Callbacks, Promises, Async/Await, Error handling

3. Building APIs with Express.js

- Routing, Middleware, Request/Response lifecycle
- RESTful API structure, CRUD operations

4. Authentication & Authorization

- JWT, bcrypt, Auth middleware, Role-based access control

5. Working with Databases

- MongoDB (Mongoose), MySQL (Sequelize/Knex)
- CRUD, Aggregation, Relationships

6. Testing & Debugging

- Unit Testing with Jest/Mocha, Postman, Supertest
- Debugging with VSCode, Error handling

7. Production-Ready Practices

- Environment variables, Logging (Winston), Rate limiting
- Helmet, CORS, Error handlers

8. DevOps & Deployment

- Docker basics, PM2, Nginx
- CI/CD pipelines, Cloud (Heroku, Vercel, AWS basics)

More to Explore:

- Redis, WebSockets, GraphQL, Microservices
- System Design, Load Balancing, Queues (Bull/Agenda)